



Mouse, the Language



[URL: http://primepuzzle.com/mouse/mouse4.html](http://primepuzzle.com/mouse/mouse4.html)

Last changed on Monday, April 23, 2007

Here's a program which tells you a holiday that's in your birth month.

~ hol.mse - ask user for birth month; give them a holiday in that month -
~ lrb - 3/17/2006 3/16/2007 - got rid of use of &aug& feature (which is
~ non-standard Mouse)

```
"What's your birth month?!"
"(Jan, Feb, Mar, Apr, May, Jun, Jul, Aug, Sep, Oct, Nov, Dec) : "
?' a: ?' b: ?' c: "!"
#L,a; #L,b; #L,c;
#H,a.,b.,c.;
$H ~ display holiday that's in 1%2%3%
3%2%1%'j=[ 'a=[ 'n=["Martin Luther King Jr. Day" @]]]
3%2%1%'f=[ 'e=[ 'b=["Ground Hog's Day" @]]]
3%2%1%'m=[ 'a=[ 'r=["Saint Patrick's Day" @]]]
3%2%1%'a=[ 'p=[ 'r=["Good Friday" @]]]
3%2%1%'m=[ 'a=[ 'y=["Cinco de mayo" @]]]
3%2%1%'j=[ 'u=[ 'n=["Father's Day" @]]]
3%2%1%'j=[ 'u=[ 'l=["Independence Day" @]]]
3%2%1%'a=[ 'u=[ 'g=["National Watermelon Day" @]]]
3%2%1%'s=[ 'e=[ 'p=["Labor Day" @]]]
3%2%1%'o=[ 'c=[ 't=["Halloween" @]]]
3%2%1%'n=[ 'o=[ 'v=["Thanksgiving Day" @]]]
3%2%1%'d=[ 'e=[ 'c=["Hanukah" @]]]
"Month not recognized" '!!' @
$L ~ lower-case what's at 1%
'a 1%.>[1%. 32+ 1%:] @
```

1. The if statement in Mouse uses the left and right brackets ([and]). When a left bracket is detected, the Stack is popped and if the popped value is positive, the statements inside the brackets are executed. If the popped value is zero or negative, the statements following the matching right bracket are executed.

2. The two character command '?' reads a single character from the keyboard.

3. Here's our first example of a macro that uses parameters. Notice how we grab the three letters in the birth month and pass their addresses to the L macro which lower-

cases the characters if needed. We then pass the three characters to the H macro to see which holiday to display. As soon as we get a hit, we exit the macro. If we never get a hit, we tell the user we don't recognize their input.

```
B0>mouse-p
MOUSE, Peter Grogono, 1983; Lee Bradley, 2007
Enter Mouse program's name : hol.mse
What's your birth month?
(Jan, Feb, Mar, Apr, May, Jun, Jul, Aug, Sep, Oct, Nov, Dec) : apr
Good Friday
B0>
```

4. Shouldn't we exit with a "Month not recognized" if we find, after determining we have "ja," that we don't have "n?" How about "ju" but neither "n" nor "l?" Other months that start with the same letter (Mar and May, Apr and Aug) have similar issues.

Here's a more efficient, somewhat harder to follow, version. I had to use Mouse's trace feature, which you'll learn about later, to get this to work. Notice how we have to push the third and sometimes the second parameter back onto Stack for these special months so we can retest them because they get popped by an if; Mouse does not have an else so we have to work a little harder!

```
~ holi.mse - lrb - 4/23/2007
"What's your birth month?!"
"(Jan, Feb, Mar, Apr, May, Jun, Jul, Aug, Sep, Oct, Nov, Dec) : "
?' a: ?' b: ?' c: "!"
#L,a; #L,b; #L,c;
#H,a.,b.,c.;
$H ~ display holiday that's in 1%2%3%
3%2%1% 'j=['a=['n=["Martin Luther King Jr. Day" @]#B;@]
      3%2% 'u=['n=["Father's Day" @]
          3% 'l=["Independence Day" @]#B;@]#B;@]
3%2%1% 'm=['a=['r=["Saint Patrick's Day" @]
      3% 'y=["Cinco de mayo" @]#B;@]#B;@]
3%2%1% 'a=['p=['r=["Good Friday" @]#B;@]
      3%2% 'u=['g=["National Watermelon Day" @]
3%2%1% 'f=['e=['b=["Ground Hog's Day" @]#B;@]#B;@]
3%2%1% 's=['e=['p=["Labor Day" @]#B;@]#B;@]
3%2%1% 'o=['c=['t=["Halloween" @]#B;@]#B;@]
3%2%1% 'n=['o=['v=["Thanksgiving Day" @]#B;@]#B;@]
3%2%1% 'd=['e=['c=["Hanukah" @]#B;@]#B;@]
#B;@
$B"Month not recognized" '!!!' @
$L ~ lower-case what's at 1%
'a 1%.>[1%. 32+ 1%:] @
```

- 0 - Emulator, Interpreter and the classic Hello, World! program
- 1 - Output string, output character, new line, Mouse's Stack, push character, end program
- 2 - Loops, variables, assignment, evaluation, comparison, comments, arithmetic, input number, output number
- 3 - Introduction to defining and calling macros
- 4 - If, input character, more on macros
- 5 - Recursive macros, the remainder operator, the small, whole number limitation of Mouse
- 6 - A few remarks about CP/M and MYZ80
- 7 - Testing and the trace feature, contact information